

Spatiotemporal Consistency

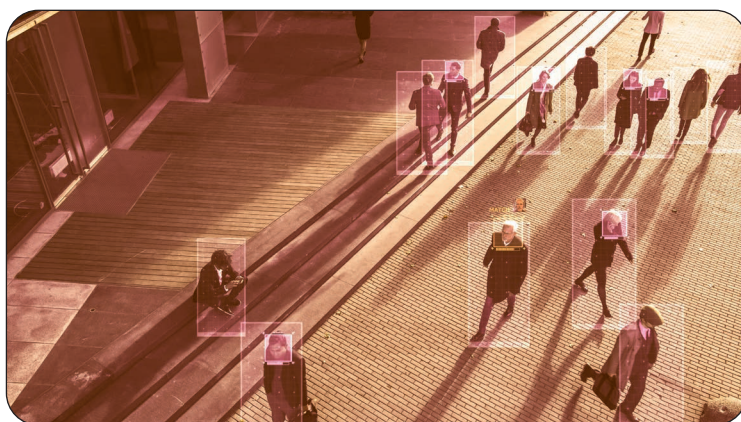
A Universal Defense Against Attacks on Autonomous Systems

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This article examines spatiotemporal consistency-based defenses, showcasing detectable spatiotemporal and contextual anomalies. We explore the real-world applications, current limitations, and future research directions of these spatiotemporal consistency-based defenses, emphasizing their transformative potential for securing autonomous systems across diverse domains.

Autonomous systems, integral to domains from autonomous driving to surveillance, enhance efficiency but also introduce significant vulnerabilities to increasingly sophisticated adversarial attacks. These attacks can disrupt critical operations, compromise safety, and erode public trust, underscoring the urgent need for robust defenses

Over the years, researchers have proposed numerous defenses to counter these attacks. Among these, methods leveraging the principles of spatiotemporal consistency (an object's own position in space and movement over time)



and contextual consistency (an object's relationship with its environment and other entities) have gained traction due to their ability to detect anomalies. For example, a stop sign appearing to float mid-air and move at an impossible

velocity would violate spatiotemporal consistency, as its position and motion are physically inconsistent. If this sign was also detected in the middle of a high-speed freeway, it would additionally breach contextual consistency, as its presence is

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incongruous with that specific traffic environment. While spatiotemporal reasoning has been applied in other domains (e.g., control systems using model-based approaches), the unique challenges posed by adversarial attacks on machine language (ML)-based perception, such as the complexity of threat models and the diversity of attack strategies, have spurred significant innovation in adapting spatiotemporal consistency techniques for autonomous systems.

In this article, we delve into the current landscape of attacks on autonomous systems and explore how spatiotemporal consistency-based defenses address them. We begin by outlining the state-of-the-art attacks against these systems, highlighting their shared traits and the inconsistencies they introduce. We then explain how spatiotemporal defenses leverage these inconsistencies to detect attacks effectively. Finally, we examine the limitations of existing defenses and propose directions for future research to enhance their robustness against evolving threats.

Perception Attacks

Autonomous systems rely on a pipeline consisting of sensors, perception stack (composed of multiple modules like object detection and object tracking), and decision-making algorithms to operate effectively in dynamic environments. This pipeline forms the backbone of the system's ability to perceive its surroundings and make real-time decisions. However, their reliance on ML models and sensor data makes them vulnerable to a variety of adversarial attacks. These attacks exploit weaknesses in perception pipelines, which can lead to catastrophic consequences, especially in safety-critical domains such as autonomous driving and surveillance.

This section explores the fundamental types of perception attacks that target autonomous systems. The most prominent attacks in current literature fall into a few key categories: misclassification, object creation/deletion, tracker hijacking, and sensor spoofing. Each type of attack undermines the integrity of the perception module in unique ways, often with severe implications for the system's overall functionality and safety.

Misclassification Attacks

Misclassification attacks manipulate a perception module to incorrectly identify objects, often using subtle adversarial perturbations or physical patches. For instance, as illustrated in Figure 1, researchers demonstrated that an adversarial patch could cause a stop sign to be misclassified as a speed limit sign, leading to dangerous driving decisions like speeding through an intersection. Such attacks undermine system planning and can cause unsafe behaviors in vehicles or bypassed detection in surveillance systems.

Object Deletion Attacks

As shown in Figure 2, object deletion attacks remove critical objects from the perception system's view. An attacker achieves this by using adversarial noise to disrupt camera feeds or lidar spoofing techniques to obscure real objects from detection. In safety-critical scenarios like autonomous driving, failing to detect pedestrians or other vehicles can result in collisions, while in industrial robotics, missing obstacles could lead to equipment damage or worker injuries.

Tracker Hijacking

Tracker hijacking manipulates the perceived trajectory of objects over time, as shown in Figure 3. Attackers can either create the illusion of a new threat, such as making a pedestrian on the sidewalk appear to veer

into traffic (a "move-in" attack), or remove a real one, such as making a vehicle in the car's path appear to have moved aside (a "move-out" attack). By disrupting temporal consistency, these attacks can cause systems to brake unnecessarily or, more dangerously, fail to avoid a collision.



Figure 1. A typical misclassification attack, where a stop sign is misclassified as a speed limit sign to get an autonomous vehicle to speed through an intersection.



Figure 2. Object deletion attacks can hide attackers attempting to break in to an area guarded by automated surveillance.

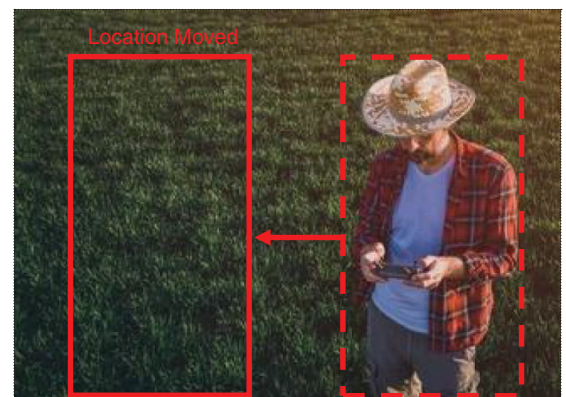


Figure 3. Tracker hijacking can cause a system to incorrectly think an object is in motion, which can change landing targets for an autonomous aerial vehicle.

Sensor Spoofing

Sensor spoofing targets the raw data from sensors before it is processed. This type of attack involves injecting false signals into GPS, lidar, radar, or cameras. For example, an attacker can manipulate GPS signals to make a vehicle believe it is in a different location, send timed laser

pulses to create false reflections in a lidar unit, or project deceptive images into a camera's field of view.

A Common Thread

Although attacks on autonomous systems vary widely in their methods and objectives, they share underlying traits that disrupt the coherence of perception data. These disruptions create detectable inconsistencies related to how objects behave in space and time (spatiotemporal consistency), and how they relate to their surrounding scene and other entities (contextual consistency). For example, Figure 4 illustrates a sensor spoofing attack where a stop sign is injected into an image. This injected object exhibits multiple inconsistencies: It appears to be floating and moving at an impossibly high velocity (spatiotemporal), and it is shown in the middle of a freeway (contextual). Defenses can leverage these spatiotemporal and contextual inconsistencies to identify adversarial behavior. This section explores these common inconsistencies and highlights the challenges posed by adaptive attacks, which attempt to exploit these defenses.

First, *spatiotemporal inconsistencies* arise when an object's observed attributes or movement violate logical or physical expectations. Attacks can generate impossible object positions, such as an object floating in the air, or induce implausible movement, like a vehicle suddenly teleporting. For example, a misclassification attack might result in an object classified as a "pedestrian" moving at 70 mph, which is inconsistent with its label.

Next, *contextual inconsistencies* occur when an object or its behavior is incongruous with its surroundings. Because attackers typically target only a small subset of objects in a scene, the environmental context provides clues for detection. A stop sign appearing on a high-speed freeway or a vehicle

detected inside a building are examples of contextual inconsistencies that can indicate an attack. Returning to the example in Figure 4, an injected stop sign that is floating, moving at high velocity, and in the middle of a freeway exhibits spatial, temporal, and contextual inconsistencies simultaneously.

Spatiotemporal Defenses

Spatiotemporal consistency-based defenses have emerged as a powerful way to safeguard autonomous systems by exploiting the inherent coherence of real-world environments to detect anomalies. By modeling spatial relationships, temporal dynamics, and contextual consistency, these methods provide a robust framework for detecting attacks that target perception modules.

Spatiotemporal Modeling

Spatiotemporal modeling lies at the heart of these defenses, using techniques like recurrent neural networks and Kalman filters to analyze object trajectories and motion patterns. By maintaining a history of observed behaviors, these models can detect abrupt changes introduced by attacks. For instance, if a tracker hijacking attack causes a vehicle to suddenly "jump" into a lane, a spatiotemporal model can flag the trajectory as anomalous.

Cross-Sensor Validation

Since autonomous systems often rely on multiple sensors like cameras, lidar, and radar, cross-sensor validation can detect spatial inconsistencies introduced by attacks by comparing data from these different sources. A spoofing attack, for example, might create false lidar reflections that do not correspond to any object detected by the camera. By correlating data across sensors, defenses can identify discrepancies that indicate tampering, with one recent work⁵ proposing



Figure 4. This sensor spoofing attack that injects a stop sign into the image has all three inconsistencies: The stop sign is floating in the air (spatial), moving at an absurd velocity (temporal), and appears in the middle of a freeway (contextual).

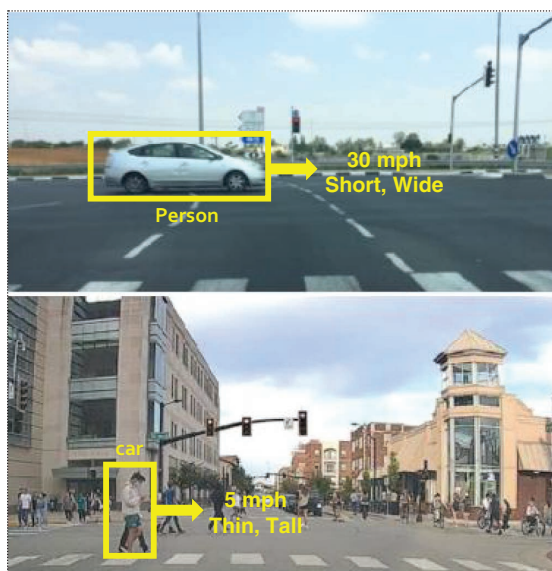


Figure 5. An example of how context can provide features for a defense. A misclassification attack turns the car (top) into a pedestrian, and the pedestrian (bottom) into a car. However, the *context* of the scene preserves clues of the original object identities: The supposed "person" is traveling at high speeds and is short and wide, while the supposed "car" is traveling very slowly across a pedestrian crosswalk, and is thin and tall. These are all features that spatiotemporal consistency-based defenses take advantage of to detect attacks.

the use of multiple sensors to create a 3D kinematic model of the environment for attack detection.

Contextual Consistency

Contextual reasoning adds another layer of robustness by analyzing the relationships between objects and their environment. As shown in Figure 5, even if a misclassification attack incorrectly labels a car as a “person,” the surrounding context preserves clues of its true identity. Features such as the object’s high speed, its “short and wide” dimensions, and its location on a road can be used by a defense to detect the misclassification; this approach is used by several major contextual consistency approaches.^{1,5} More advanced defensive methods² also extract individual key points from an object (such as arms and legs) and compare them with the object’s location to ensure that the parts are consistent with the whole.

Dynamic Thresholding

Static thresholds for anomaly detection used in older literature can be insufficient because environmental conditions vary significantly. Dynamic thresholding addresses this by using advanced techniques to adapt detection thresholds based on real-time environmental factors. For instance, recent research has introduced agent-based dynamic thresholding,⁴ which utilizes deep neural networks to optimize thresholds in ML models, as well as hybrid spatiotemporal neural networks,³ which combine recurrent and spiking neural networks to flexibly adapt to environmental conditions. Using these advanced artificial intelligence (AI) techniques allows a system to adjust its sensitivity to account for challenging scenarios, such as high-traffic situations with a greater likelihood of occlusions.

Evading Consistency

One way for attackers to get around consistency checks is to conduct an *adaptive attack*. In an adaptive attack, the attacker knows whatever metric that the victim is using to detect attacks (e.g., contextual consistency), and they modify their attack to produce results that are contextually consistent. For example, an attacker might carefully craft perturbations that align with plausible spatial

Current Challenges

Key challenges remain for spatiotemporal defenses. A primary issue is their limited ability to generalize across different domains, such as from autonomous driving to surveillance, as spatiotemporal dynamics vary significantly. Another significant challenge is distinguishing between adversarial attacks and natural errors caused by sensor noise, occlusions, or poor

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trajectories while introducing minimal temporal deviations. As another example, projection-based tracker hijacking attacks might exploit valid surfaces and lighting conditions to minimize detectability.

However, even adaptive strategies struggle to maintain consistency across both spatial and temporal dimensions simultaneously. The inherent complexity of ensuring coherence in all aspects (e.g., spatial positioning, temporal behavior, and contextual alignment) makes these attacks challenging to execute without introducing detectable anomalies. For example, a recent defense² demonstrated that in order for an adaptive attack to evade its detection scheme, attackers would have to make an untenable amount of perturbations to the input image, which in the tested real-world locations was not achievable.

Looking Ahead

While spatiotemporal defenses show significant potential, they face limitations that must be addressed to ensure their scalability and robustness in real-world applications.

lighting. Furthermore, a critical hurdle is transitioning from manually defined consistency rules, which are labor-intensive, to methods that can automatically learn a comprehensive set of rules from data in a scalable manner.

Future Directions

Future work can improve these defenses in several ways. A critical research area is developing universal models grounded in physics AI that are capable of cross-domain generalization, perhaps using transfer learning to adapt models from one domain to another with minimal retraining. Another priority is enhancing robustness through multimodal sensing, where combining data from cameras, lidar, and radar can help cross-validate information more reliably. Finally, all defenses must be examined through the lens of real-world applicability; moving beyond simulations to test systems in various operational scenarios is essential for assessing their practical effectiveness and identifying edge cases.

This work studied the evolution of spatiotemporal consistency-based defenses for autonomous systems across multiple domains. It demonstrated that these defenses provide a powerful tool for detecting adversarial attacks by exploiting the inherent coherence in space and time, enabling the uncovering of even sophisticated attack patterns. The article explored various types of attacks, their common features, and how spatiotemporal defenses can effectively detect them. While challenges still remain, including adapting to new attack strategies and improving cross-domain generalization, ongoing research promises to make these defenses more robust and universally applicable. As spatiotemporal defenses continue to improve, they will significantly raise the bar for attackers, requiring more resources and expertise to bypass them. This evolution in defensive capabilities will ultimately contribute to making autonomous systems safer and more reliable in an increasingly interconnected world, enhancing their resilience against a wide range of adversarial threats. ■

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